

# Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

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## Abstract

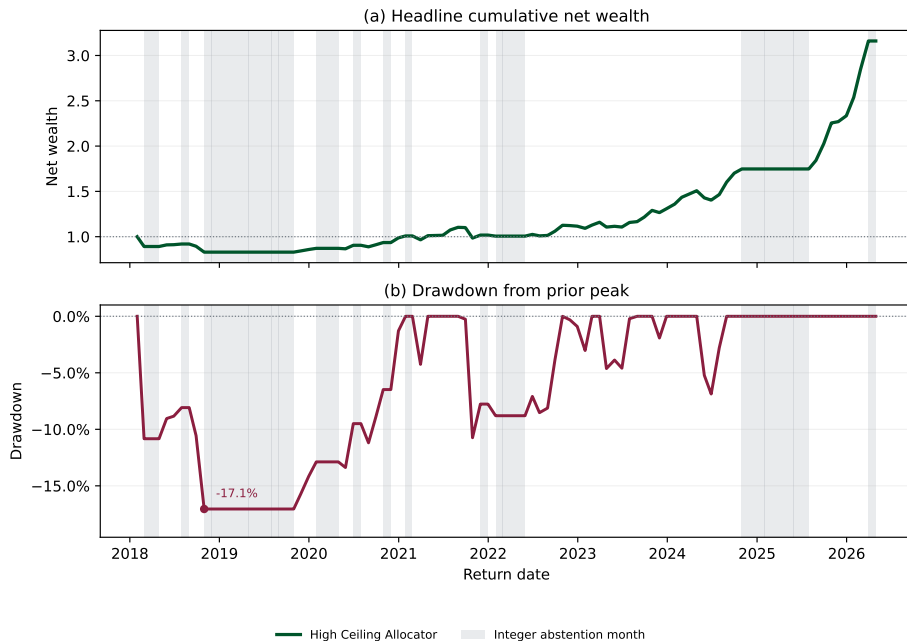
This paper presents one retrospective development result. The High Ceiling Allocator on the eight-name plus VIX universe returns 16.0 percent annualized with a Sortino ratio of 2.275 and a maximum drawdown of 17.1 percent under frozen modeled costs. Its Sortino ratio is 2.052 at observed touch fills, with 94.1 percent entry coverage in a licensed-quote audit. The allocator combines premium accounting, a complete Greek factor and residual covariance, asymmetric costs, cash, and hard survival gates. The result is not a confirmatory claim. Confirmation requires 36 untouched monthly observations beginning 2026-07-31 under frozen code, data rules, and hashes.

### Key findings and claim boundary

- The eight-name plus VIX High Ceiling Allocator is the headline development result because it survives the frozen risk and observed-quote checks.
- The Survival Allocator retains Greek factor and residual cross terms, prices predictable costs inside utility, permits cash, and rejects infeasible integer books.
- The Sharpe Prototype fails the survival standard when its VIX-enabled CPCV paths absorb at zero in March 2020.
- Every reported path remains retrospective. The prospective freeze sets the boundary for any later confirmatory claim.

## 1. Introduction and headline result

The paper asks one question. Can an option-only allocator survive costs, sparse quotes, whole-contract execution, and option-specific tail risk while retaining useful historical performance? The evidence supports only a limited answer. One frozen development



**Figure 1:** Headline retrospective development path for the High Ceiling Allocator on the eight-name plus VIX universe. Panel (a) compounds net returns under frozen modeled costs. Shading marks direct-or-abstain cash intervals. Panel (b) reports drawdown from the prior peak.

configuration survives the checks in this paper. It is not a live track record and it is not a confirmatory result.

The research path produced three mechanisms. The Sharpe Prototype chose a high historical Sharpe direction. The Survival Allocator repaired its covariance, objective, cash rule, and survival gates. The High Ceiling Allocator kept those repairs while allowing a higher risk ceiling and a direct-or-abstain whole-contract decision.<sup>1</sup>

Figure 1 shows the headline path. The shaded intervals are cash abstentions after the direct integer conversion failed a hard constraint. Cash is an allowed portfolio, not a repaired risky book. The drawdown panel keeps the claim tied to the path rather than to one ratio.

The portfolio problem begins with mean-variance choice (Markowitz, 1952; Sharpe, 1966). Listed options require different accounting because premium, collateral, and loss exposure are not the same object. Black’s futures option model supplies the traded forward anchor for VIX options (Black, 1976). Existing option-allocation work already combines option returns, Greeks, and portfolio choice (Faia and Santa-Clara, 2017; Zhao

<sup>1</sup>Artifact files and machine-readable ledgers retain the internal codes E1 (Sharpe Prototype), R1 (Survival Allocator), and R1.1 (High Ceiling Allocator). The internal codes remain in artifact filenames and verifier keys so the frozen record stays reproducible.

and Palomar, 2018; Malamud, 2014).

The expected-return channels also have established foundations. Coval and Shumway link expected option returns to option exposure (Coval and Shumway, 2001). Delta-hedged option returns reveal compensation for volatility risk (Bakshi and Kapadia, 2003). The variance risk premium predicts aggregate returns (Bollerslev et al., 2009). The validation design follows the literature on backtest overfitting, multiple testing, and combinatorial cross-validation (White, 2000; Hansen, 2005; Bailey et al., 2017; Bailey and López de Prado, 2014). The contribution here is an auditable implementation and a clear record of where earlier implementations fail.

## 2. Theory in brief

### 2.1. Premium accounting

An option position is a funded cashflow. Let  $C_{i,t}$  be the option mark,  $m_i$  its multiplier,  $N_{i,t}$  the signed contract count, and  $W_t$  portfolio NAV. Define *premium weights* by

$$w_{i,t} = \frac{N_{i,t}m_iC_{i,t}}{W_t}.$$

If  $r_{i,t+1} = (C_{i,t+1} - C_{i,t})/C_{i,t}$ , the option contribution to the NAV return is

$$\frac{\Delta W_{i,t+1}}{W_t} = w_{i,t}r_{i,t+1}.$$

The multiplier cancels because the weight is already premium normalized. A negative weight records premium received but does not cap the later short-option liability. Gross premium, margin, collateral, stress loss, and liquidity must therefore remain separate controls. Residual NAV stays in cash.

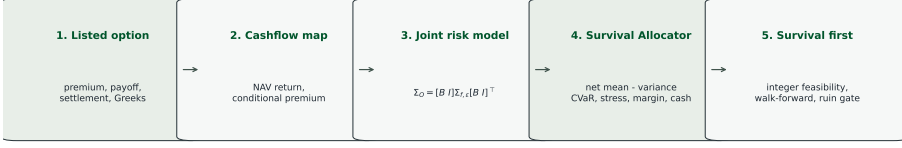
### 2.2. Greek-induced covariance

For each contract, the local return map uses premium-normalized delta, centered quadratic spot exposure, and vega. Stack the state innovations in  $f_{t+1}$ , the fixed Greek loadings in  $B_t^G$ , and the unexplained option return in  $\varepsilon_{t+1}$ . Then

$$r_{O,t+1} = \mu_{O,t} + B_t^G f_{t+1} + \varepsilon_{t+1}.$$

Write  $\Omega_t = \text{Var}_t(f_{t+1})$ ,  $\Gamma_t = \text{Cov}_t(f_{t+1}, \varepsilon_{t+1})$ , and  $\Sigma_{\varepsilon,t} = \text{Var}_t(\varepsilon_{t+1})$ . The option covariance is

$$\Sigma_{O,t} = B_t^G \Omega_t (B_t^G)^\top + B_t^G \Gamma_t + \Gamma_t^\top (B_t^G)^\top + \Sigma_{\varepsilon,t}.$$



**Figure 2:** Theory and implementation flow. Listed contracts become NAV-normalized cash-flows. The joint factor and residual covariance enters cost-aware net utility. Hard survival and execution gates apply to the whole-contract book.

The two cross terms generally remain because the code fixes  $B_t^G$  from Greeks. It does not estimate an ordinary least-squares projection that would force orthogonal residuals. The implementation estimates the joint covariance of factors and residuals, then applies the linear map  $[B_t^G \ I]$ . This construction preserves positive semidefiniteness and the observed cross block.

For VIX options, the state anchor is the matched VX futures contract rather than equity spot. VIX and VVIX can describe the state, but they do not replace the traded forward in the option return map. Settlement evidence must also follow the listed contract rule.

### 2.3. Expected return and net utility

The structural mean routes compensation through the same economic exposures:

$$\hat{\mu}_{i,t} = b_{i,t}^\Delta \hat{\lambda}_t^S + b_{i,t}^\Gamma \hat{\lambda}_t^{var} + b_{i,t}^\nu \hat{\lambda}_t^{vol} + b_{i,t}^{skew} \hat{\lambda}_t^{skew} + b_{i,t}^{tail} \hat{\lambda}_t^{tail} + \theta_{i,t}^*.$$

Skew and tail terms enter the mean and named stress tests. They are not covariance loadings. The separation avoids pretending that the available panel identifies a larger Greek exposure map.

The Survival and High Ceiling Allocators solve one signed-position problem. For positive and negative parts  $w_+$  and  $w_-$ , the objective is

$$\max_{w \in \mathcal{C}} \left\{ \hat{\mu}_t^\top w - c_{+,t}^\top w_+ - c_{-,t}^\top w_- - \frac{\gamma}{2} w^\top \Sigma_{O,t} w \right\}.$$

The asymmetric cost vectors sit inside the objective. The feasible set permits  $w = 0$  and limits gross premium, short premium, volatility, CVaR, named stress loss, margin, collateral, assignment exposure, Greek concentration, and point-in-time liquidity. A whole-contract book receives a score only after the implementation checks every gate again. An infeasible book fails closed to cash.

Figure 2 separates modeling from acceptance. A favorable mean or variance estimate cannot override a zero-wealth path, a margin breach, or an infeasible integer conversion. This ordering is the central repair.

### 3. Data, universes, and cost model

The empirical panel contains listed equity options, underlying state variables, and the matched volatility instruments needed by the four frozen universes. Every rebalance uses only information available by its decision date. Table 1 records the four comparison sets and their frozen spread inputs.

**Table 1:** Frozen empirical universes and baseline evidence. Prototype net is the static full-cost net Sharpe ratio used to describe the earlier development experiment.

| Config     | Universe          | Spread input                           | Status              | Prototype net | Naive net | Stock Markowitz |
|------------|-------------------|----------------------------------------|---------------------|---------------|-----------|-----------------|
| orig       | 8 equities        | exact panel CBBO                       | capacity diagnostic | 0.778         | 0.365     | 0.909           |
| orig+VIX   | 8 equities + VIX  | exact panel CBBO + inferred CBBO proxy | research candidate  | 1.383         | -0.890    | 0.909           |
| larger     | 56 equities       | exact panel CBBO + inferred CBBO proxy | mixed               | 0.587         | 0.596     | 0.909           |
| larger+VIX | 56 equities + VIX | exact panel CBBO + inferred CBBO proxy | research candidate  | 1.628         | 0.293     | 0.909           |

The original universes contain eight equity names. The larger universes contain 56. The design evaluates each set with and without VIX options. Contract selection applies maturity, moneyness, mark, open-interest, volume, and quote-quality filters before optimization. The same point-in-time panel supplies marks, Greeks, and realized option returns.

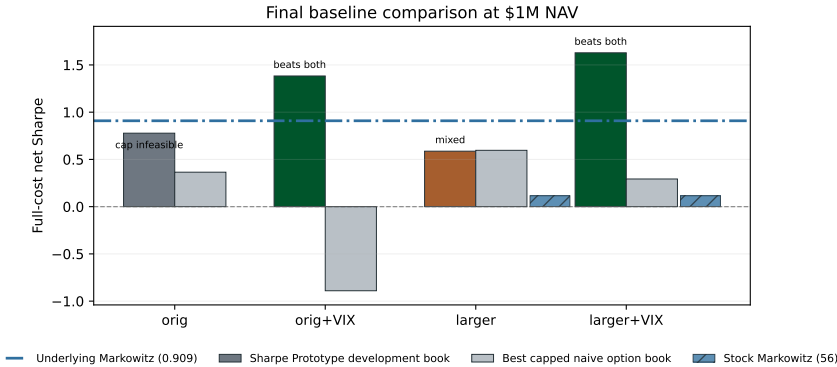
The cost stack includes half-spread, fees, direction-specific slippage, short-call borrow, and funding where applicable. These transaction costs enter both selection and reported net returns. The stack is asymmetric because buying at the ask and covering a short at the ask are not mirror images of receiving the bid.

Spread evidence follows a fixed ladder. The eight equity names use exact historical panel CBBO. Rows outside that panel use an inferred CBBO proxy when no matched historical quote exists. Broad-name and VIX proxy rows are execution sensitivity, not execution truth. The licensed-quote audit later tests that distinction directly. VIX settlement requires the appropriate listed settlement record rather than a convenient close.

The baseline comparison includes ordinary underlying Markowitz and capped naive option books. The stock baseline asks whether option complexity earns its place. The capped naive book asks whether estimation adds value beyond simple option diversification under the same costs and capacity rules.

4. The Sharpe Prototype and its failure

The Sharpe Prototype maximized a cost-adjusted Sharpe direction after inspecting the post-2020 development window. That design can rank attractive paths, but it does not make the path selection prospective. Figure 3 and Table 2 retain the result as a record of the research path.



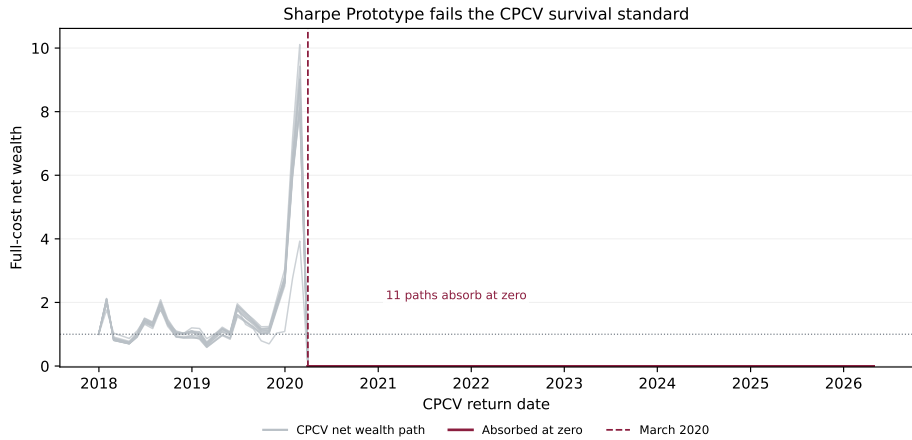
**Figure 3:** Static full-cost net comparison for the Sharpe Prototype, capped naive option books, and stock baselines at the frozen account size. The exhibit is retrospective development evidence.

**Table 2:** Structural-mean channel ablation for the Sharpe Prototype. Each row re-solves and re-scores the book after removing one channel.

| Arm               | orig  | orig+VIX | larger | larger+VIX |
|-------------------|-------|----------|--------|------------|
| Full model        | 0.778 | 1.383    | 0.587  | 1.628      |
| No carry          | 0.774 | 0.463    | 0.574  | 0.483      |
| No delta          | 0.370 | 1.479    | 0.672  | 1.684      |
| No vol/VRP        | 0.772 | 1.382    | 0.587  | 1.600      |
| No skew/tail      | 0.772 | 1.372    | 0.586  | 1.629      |
| No relative value | 0.822 | 1.355    | 0.504  | 1.570      |

The liquid-era CPCV paths reveal the survival failure. Figure 4 pivots the artifact to one column per path and compounds each path independently. Every VIX-enabled path reaches the absorption boundary in March 2020. Arithmetic return ratios after that event cannot restore wealth.

This is a survival failure. It also explains why the prototype cannot anchor the paper even when a later window shows large compounding. Researchers selected the model



**Figure 4:** Full-cost CPCV wealth paths for the VIX-enabled Sharpe Prototype. All paths absorb at zero in March 2020. The maroon segment marks the absorbing state.

after inspecting that later window. The window omits the event that destroys the earlier paths.

5. The Survival Allocator

The Survival Allocator repairs the earlier objective and acceptance rule. It retains the factor and residual covariance cross terms. It prices predictable costs inside utility. It allows cash and removes offsetting split-leg gross. It checks CVaR, stress, margin, collateral, assignment, liquidity, and integer feasibility before accepting a position.

**Table 3:** Survival Allocator monthly walk-forward development summary on the common post-2020 window. Terminal wealth and tail statistics describe the whole-contract book.

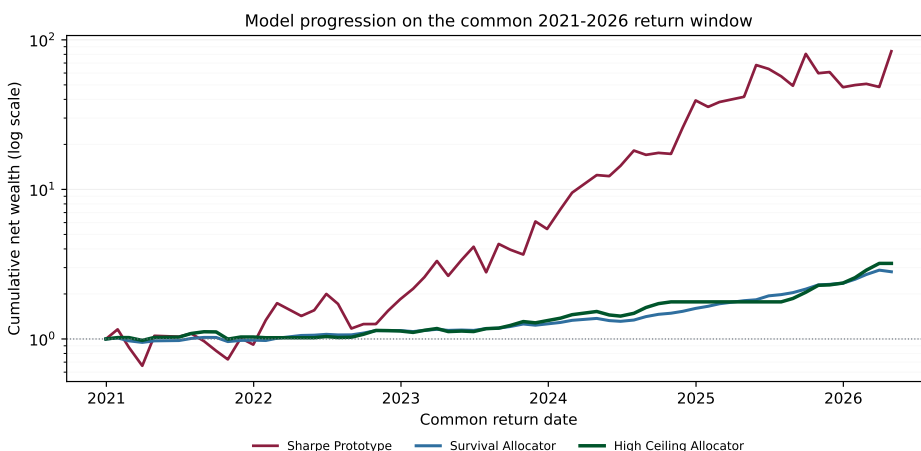
| Universe   | Obs. | Mean gross | Terminal wealth | Ann. geom. | Worst month | ES <sub>95</sub> | Verdict              |
|------------|------|------------|-----------------|------------|-------------|------------------|----------------------|
| orig+VIX   | 60   | 0.049      | 2.814           | 0.230      | -0.062      | -0.046           | development_survived |
| larger+VIX | 60   | 0.050      | 2.401           | 0.191      | -0.059      | -0.048           | development_survived |
| orig       | 60   | 0.032      | 1.220           | 0.041      | -0.078      | -0.067           | development_survived |
| larger     | 60   | 0.038      | 1.180           | 0.034      | -0.070      | -0.061           | development_survived |

Table 3 shows that all four frozen universes pass the development survival checks. The 15 percent predicted-volatility ceiling also limits return potential. That ceiling is a deliberate constraint, not an optimization error. It motivates a separately frozen higher-ceiling development policy while preserving the survival-first architecture.

## 6. The High Ceiling Allocator

The High Ceiling Allocator raises the predicted-volatility ceiling to 25 percent. It pursues a 50 percent gross target only along positive-edge directions that the cost-aware first stage selects. It never forces that target through an infeasible book. The continuous target converts directly to whole contracts by truncating toward cash. A feasible conversion trades. An infeasible conversion abstains.

The comparison in Figure 5 needs an explicit credibility warning. The prototype's large line begins after its March 2020 zero-absorption failure, so the common window excludes the event that invalidates it. Researchers also selected the prototype after inspecting that window. The inferred CBBO proxy supplies its broad spread evidence. The High Ceiling line is smaller because it belongs to the frozen survival-first policy that the paper can honestly defend.

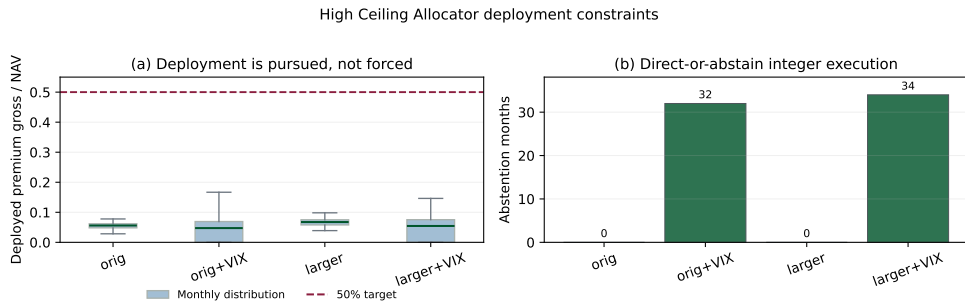


**Figure 5:** Common-window net wealth on a logarithmic scale. The Sharpe Prototype compounds to roughly 84 times wealth while the High Ceiling Allocator reaches about 3.2 times. The larger prototype number is not a credible headline. This window starts after March 2020 and excludes the prototype's CPCV zero-absorption failure. Researchers selected the prototype after inspecting the window. The inferred CBBO proxy supplies its broad spread evidence. The smaller High Ceiling result is the honest headline because it follows the frozen survival-first policy.

Figure 6 shows why the paper describes the target as pursued. The observed gross distributions sit well below the reference line. The VIX-enabled books sometimes choose cash when direct whole-contract conversion is infeasible.

Table 4 keeps the base replay separate from the EGARCH diagnostic.

The VIX-40 rule reacts only after an official close and uses the first executable quote



**Figure 6:** Deployment and integer execution for the High Ceiling Allocator. Panel (a) compares realized gross premium with the target. Panel (b) counts cash abstentions under the direct-or-abstain rule.

**Table 4:** Chronological High Ceiling development replay. Target hit is the share of whole-contract books at or above the gross target. EGARCH remains diagnostic.

| Universe   | Arm              | Obs. | Mean gross | Target hit | Terminal | ES <sub>95</sub> | Verdict              |
|------------|------------------|------|------------|------------|----------|------------------|----------------------|
| larger     | EGARCH           | 93   | 0.075      | 0.00       | 2.736    | -0.113           | development_survived |
| larger     | High Ceiling 25% | 93   | 0.067      | 0.00       | 2.454    | -0.098           | development_survived |
| larger+VIX | EGARCH           | 93   | 0.053      | 0.00       | 2.758    | -0.076           | development_survived |
| larger+VIX | High Ceiling 25% | 93   | 0.047      | 0.00       | 2.804    | -0.067           | development_survived |
| orig       | EGARCH           | 93   | 0.062      | 0.00       | 2.418    | -0.105           | development_survived |
| orig       | High Ceiling 25% | 93   | 0.055      | 0.00       | 2.368    | -0.100           | development_survived |
| orig+VIX   | EGARCH           | 93   | 0.058      | 0.00       | 3.018    | -0.076           | development_survived |
| orig+VIX   | High Ceiling 25% | 93   | 0.048      | 0.00       | 3.160    | -0.077           | development_survived |

after the next regular-session open. The paper leaves the event return unscored because the frozen repository lacks complete matched executable quotes for every held option. The rule does not borrow the no-rule return.

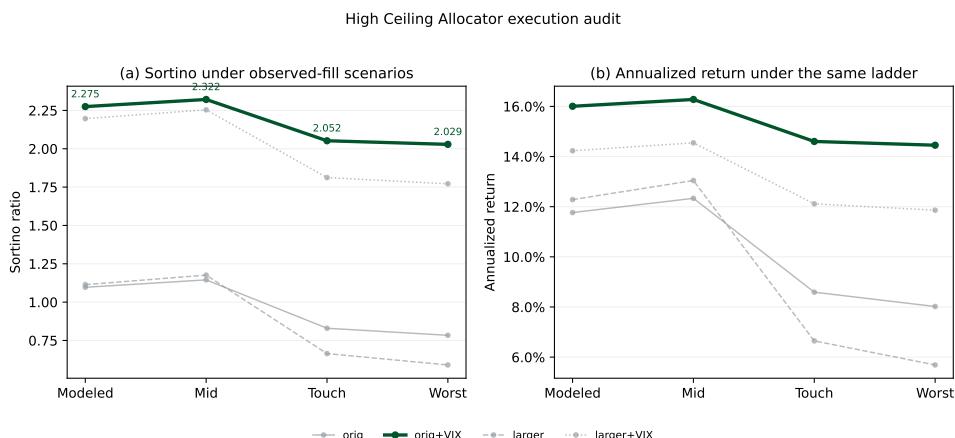
The EGARCH overlay also remains outside the candidate. It achieved broad forecast coverage but failed its prespecified forecast-loss and tail-risk gate. A diagnostic that fails promotion cannot improve the headline path by reinterpretation.

## 7. Execution audit at observed quotes

The licensed-quote audit changes only spread evidence. It preserves positions, returns, non-spread costs, integer decisions, and survival gates. The mid scenario uses observed midpoints while retaining the frozen non-spread terms. The touch scenario substitutes observed entry and exit spreads. The worst scenario uses the entry-window maximum spread and the observed exit spread.

Months without observed quotes keep the frozen modeled cost. The audit counts them and never imputes them. Reconstructing modeled cost from the frozen components leaves

a maximum absolute gap below  $10^{-16}$ . The scenario ladder therefore isolates the effect of quote-based spread evidence.



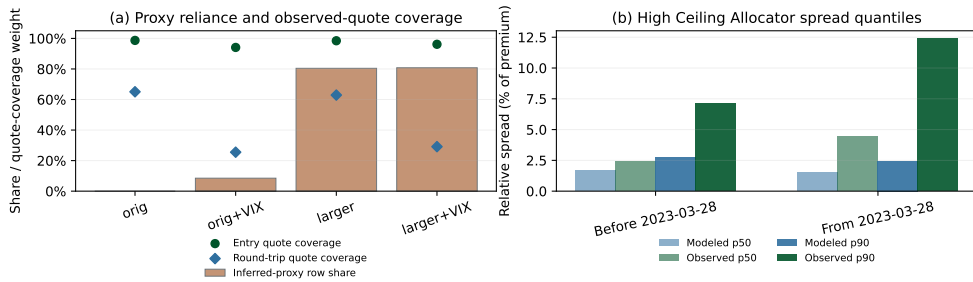
**Figure 7:** Execution audit for the High Ceiling Allocator. The x axis moves from frozen modeled costs through observed mid, touch, and worst fills. The eight-name plus VIX configuration uses the heavier line.

Figure 7 shows a modest improvement at observed mid and a decline at touch and worst fills. The headline configuration remains above its survival threshold. The ladder is an execution sensitivity around the same frozen positions. It is not a new backtest selected from the quote sample.

### 7.1. Why the eight-name book with VIX is the headline

The spread-source distinction decides which historical result deserves emphasis. Broad-name and VIX rows lack exact panel CBBO in the frozen cost stack, so those rows use the inferred proxy. The licensed audit finds wider observed broad-name spreads than the proxy assumed. Figure 8 places proxy reliance, observed-quote coverage, and spread quantiles in one view.

For larger+VIX, touch Sortino falls from 2.197 to 1.812, a 17.5 percent decline. For orig+VIX, it falls from 2.275 to 2.052, a 9.8 percent decline. Round-trip coverage is 29.1 percent for the broad book against 65.1 percent for the eight-name book without VIX. The headline therefore uses the eight-name plus VIX configuration because its equity-option spread evidence is exact in the frozen panel, its entry coverage is high, and its observed touch result remains viable. Appendix A.4 reports the wide scenario, spread, and liquidity tables.



**Figure 8:** Proxy reliance and licensed-quote evidence. Panel (a) compares inferred-proxy row share with absolute-weighted entry and round-trip quote coverage. Panel (b) pools the High Ceiling positions across configurations and compares modeled and observed spread quantiles before and after the quote-schema cutover.

## 8. Robustness in brief

The validation suite separates path fragility from formal inference. CPCV recombines purged test blocks and preserves the zero-wealth boundary.

Monte Carlo resampling tests return ordering. Refit and reprice exercises repeat estimation and option repricing under locked cost rules. These simulations serve as path-risk diagnostics only, not as tradable histories or confirmatory observations.

Figure 9 makes the asymmetry visible. The original-universe repricing checks produce mixed evidence, while the broad-universe overlays fail. The estimated probability of backtest overfitting is about 0.19. That value cautions against choosing a winner from many related variants even when a static ratio looks strong.

Rolling out-of-sample tests retrain only on earlier observations. The common ledger keeps the comparison dates fixed across universes. Table 5 shows that multiple-testing adjustment weakens the interpretation of the static prototype results but does not erase every signal. The survival failure still dominates the prototype's attractive later-window statistics.

Robustness summary by test

|            |            |          |                  |                |              |          |             |          |                  |
|------------|------------|----------|------------------|----------------|--------------|----------|-------------|----------|------------------|
| orig       | diagnostic | fail     | pass             | pass           | pass         | pass     | pass        | pass     | mixed            |
| orig+VIX   | pass       | pass     | pass             | pass           | pass         | pass     | pass        | pass     | mixed            |
| larger     | mixed      | pass     | pass             | pass           | mixed        | pass     | pass        | mixed    | fail             |
| larger+VIX | pass       | pass     | pass             | pass           | pass         | pass     | pass        | mixed    | fail             |
|            | Baselines  | Capacity | CPCV net (2018+) | CPCV net claim | MC resampled | MC refit | Rolling OOS | PBO rank | Repriced overlay |

**Figure 9:** Artifact-backed robustness summary for the four frozen prototype universes. Cells distinguish passes, mixed evidence, failures, and capacity diagnostics across the locked validation suite.

**Table 5:** Formal inference for the static full-cost net prototype books over the common test window. The table reports uncertainty, probabilistic and deflated Sharpe measures, and comparisons with stock and capped naive baselines.

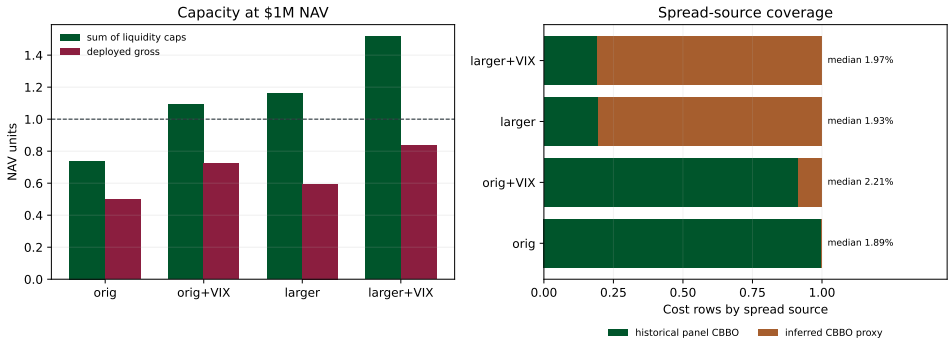
| Config     | Net Sharpe | CI lo | CI hi | PSR   | DSR   | dSR stock | p stock | dSR naive | p naive |
|------------|------------|-------|-------|-------|-------|-----------|---------|-----------|---------|
| orig       | 0.778      | 0.163 | 1.433 | 0.973 | 0.945 | -0.131    | 0.776   | 0.413     | 0.014   |
| orig+VIX   | 1.383      | 0.895 | 1.971 | 1.000 | 0.997 | 0.474     | 0.383   | 2.273     | 0.000   |
| larger     | 0.587      | 0.029 | 1.173 | 0.925 | 0.745 | -0.322    | 0.436   | -0.009    | 0.967   |
| larger+VIX | 1.628      | 1.111 | 2.245 | 1.000 | 0.996 | 0.719     | 0.179   | 1.335     | 0.000   |

## 9. Capacity, execution limits, and claim boundary

Capacity is part of the result. Figure 10 compares deployed gross with the sum of contract liquidity caps and also records the spread-source mix. The local contract panel does not support mechanical scaling from roughly \$1 million to \$5 million NAV.

Multi-exchange routing could change available depth. The paper marks routing improvement as not claimed. The data contain no routed orders or realized fills. The allocator checks assignment exposure before execution. Any whole-contract breach fails closed to cash. The paper also leaves the VIX-40 risk-off arm unscored without complete event quotes. A model price cannot fill any of these gaps after the fact.

The evidence boundary is therefore narrow. The paper supports one retrospective development candidate at the tested size and under the frozen source hierarchy. It does not support unrestricted capacity, live execution quality, or performance outside the recorded universes.



**Figure 10:** Capacity and spread-source evidence at the frozen account size. Panel (a) compares deployed gross with the sum of liquidity caps. Panel (b) shows exact panel CBBO and inferred-proxy shares.

### 10. Freeze protocol and conclusion

The prospective protocol freezes code, configuration, source hashes, artifact hashes, and the decision rule before the system scores a future observation. Confirmation requires 36 untouched monthly observations beginning 2026-07-31. A missing quote, failed integer conversion, or survival breach keeps its original status in the ledger. No later return can change that status.

The conclusion is simple. The Sharpe Prototype found an attractive later-window path but failed the March 2020 survival test. The Survival Allocator repaired the model and made cash admissible. The High Ceiling Allocator preserved those repairs and produced the one development path that remains credible after the licensed-quote audit. That path is the paper’s candidate, not its proof.

## A. Technical details and relocated evidence

This appendix records the mathematical steps needed to audit the implementation and the wide exhibits that the main reading path omits. The arguments are standard. We include them to make the code-to-equation mapping explicit.

### A.1. Funded option scale and covariance

Let  $w$  denote signed premium weights and let residual cash be the numeraire. If the constraint set limits gross premium, short premium, stress loss, margin, collateral, Greek concentration, and contract liquidity, then every economically nonzero direction uses a finite budget. A direction that lies in the joint null space of every control must carry no premium, payoff, or Greek exposure. It is economically empty. This separation rules out free scale in the feasible option book.

For the joint state vector

$$Z_t = \begin{bmatrix} f_t \\ \varepsilon_t \end{bmatrix}, \quad \hat{S}_Z = \begin{bmatrix} \hat{\Omega} & \hat{\Gamma} \\ \hat{\Gamma}^\top & \hat{\Sigma}_\varepsilon \end{bmatrix},$$

the option covariance estimator is

$$\hat{\Sigma}_O = [B^G \ I] \hat{S}_Z [B^G \ I]^\top.$$

The map is positive semidefinite whenever the estimated joint covariance is positive semidefinite. Expanding the product gives both factor and residual cross terms. With no shrinkage, the construction reproduces the aligned sample option covariance before the final numerical eigenvalue floor.

### A.2. Utility monotonicity and integer execution

Define the cost-aware value

$$V(\mathcal{C}, c) = \max_{w \in \mathcal{C}} \left\{ \hat{\mu}^\top w - \frac{\gamma}{2} w^\top \Sigma w - c^\top |w| \right\}.$$

If  $\mathcal{C}_1 \subseteq \mathcal{C}_0$  and  $c_1 \geq c_0$  componentwise, then  $V(\mathcal{C}_1, c_1) \leq V(\mathcal{C}_0, c_0)$ . Every point feasible under the tighter rules also belongs to the earlier feasible set, and its cost penalty is weakly larger. The reporting direction is therefore conservative. Higher spreads or tighter execution limits cannot improve the optimized value.

A continuous premium weight implies a contract count

$$N_i = \frac{w_i W_t}{m_i C_{i,t}}.$$

The survival-first policies truncate counts toward cash and recheck the full constraint set. They never remove one risky leg to manufacture feasibility. A failed direct conversion selects cash and keeps the rejected book's diagnostics in the audit ledger.

### *A.3. Maximum Sharpe and validation geometry*

For  $\Sigma \succ 0$ , the unconstrained maximum-Sharpe direction is proportional to  $\Sigma^{-1}\mu$ . Under a convex cone  $K$ , its normalized form is

$$\max_{y \in K} y^\top \mu \quad \text{subject to} \quad y^\top \Sigma y \leq 1.$$

This reduction determines a direction but not an economically valid option scale. The premium, stress, margin, and liquidity controls supply that scale. The earlier prototype used the ratio as its selection target. The later policies use strictly concave net utility and permit zero exposure.

Purged combinatorial cross-validation separates each test label from any training label whose option holding window could overlap it. The fold ledger checks the realized calendar gap on both sides of every test block. The selection rule excludes embargoed observations. Recombined paths retain their original ordering within each test block and apply zero wealth as an absorbing state.

### *A.4. Licensed-quote audit tables*

Table 6 contains the complete scenario ladder. The audit aggregates coverage by absolute portfolio weight. Missing quote legs retain modeled spread cost, and the coverage denominator still counts them. Tables 7 and 8 report the spread and entry-date liquidity checks.

**Table 6:** High Ceiling annualized return, Sortino, drawdown, and quote coverage under the frozen modeled, mid, touch, and worst scenarios.

| Config     | modeled ann. | modeled Sortino | modeled maxDD | mid ann. | mid Sortino | mid maxDD | touch ann. | touch Sortino | touch maxDD | worst ann. | worst Sortino | worst maxDD | Entry cov. % | Round-trip cov. % |
|------------|--------------|-----------------|---------------|----------|-------------|-----------|------------|---------------|-------------|------------|---------------|-------------|--------------|-------------------|
| larger     | 0.123        | 1.114           | -0.394        | 0.130    | 1.176       | -0.391    | 0.066      | 0.664         | -0.413      | 0.057      | 0.590         | -0.418      | 98.472       | 62.902            |
| larger+VIX | 0.142        | 2.197           | -0.121        | 0.145    | 2.254       | -0.121    | 0.121      | 1.812         | -0.128      | 0.119      | 1.771         | -0.129      | 96.157       | 29.134            |
| orig       | 0.118        | 1.097           | -0.381        | 0.123    | 1.145       | -0.378    | 0.086      | 0.830         | -0.389      | 0.080      | 0.784         | -0.390      | 98.696       | 65.061            |
| orig+VIX   | 0.160        | 2.275           | -0.171        | 0.163    | 2.322       | -0.167    | 0.146      | 2.052         | -0.185      | 0.145      | 2.029         | -0.186      | 94.096       | 25.504            |

**Table 7:** Modeled and observed relative-spread quantiles by policy and quote regime.

| Arm          | Regime          | Source   | p25   | p50   | p75   | p90   |
|--------------|-----------------|----------|-------|-------|-------|-------|
| Survival     | post_2023_03_28 | modeled  | 0.012 | 0.015 | 0.017 | 0.030 |
| Survival     | post_2023_03_28 | observed | 0.028 | 0.046 | 0.078 | 0.133 |
| Survival     | pre_2023_03_28  | modeled  | 0.011 | 0.016 | 0.017 | 0.034 |
| Survival     | pre_2023_03_28  | observed | 0.013 | 0.026 | 0.049 | 0.081 |
| High Ceiling | post_2023_03_28 | modeled  | 0.011 | 0.015 | 0.016 | 0.024 |
| High Ceiling | post_2023_03_28 | observed | 0.026 | 0.044 | 0.073 | 0.124 |
| High Ceiling | pre_2023_03_28  | modeled  | 0.013 | 0.017 | 0.018 | 0.027 |
| High Ceiling | pre_2023_03_28  | observed | 0.014 | 0.024 | 0.042 | 0.071 |

**Table 8:** Entry-date participation, open-interest diagnostics, and frozen threshold breaches in the licensed-quote audit.

| Arm          | volume_sum_p25 | volume_sum_p50 | volume_sum_p75 | volume_sum_p90 | volume_max_p25 | volume_max_p50 | volume_max_p75 | volume_max_p90 | open_interest_p25 | open_interest_p50 | open_interest_p75 | open_interest_p90 | optimizer_005_breaches | volume_010_breaches | ot_002_breaches |
|--------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-------------------|-------------------|-------------------|-------------------|------------------------|---------------------|-----------------|
| Survival     | 0.001          | 0.002          | 0.008          | 0.020          | 0.002          | 0.006          | 0.020          | 0.048          | 0.000             | 0.000             | 0.001             | 0.004             | 83                     | 27                  | 34              |
| High Ceiling | 0.001          | 0.004          | 0.012          | 0.028          | 0.004          | 0.011          | 0.033          | 0.069          | 0.000             | 0.001             | 0.002             | 0.005             | 186                    | 57                  | 140             |

### A.5. Integer, concentration, and validation tables

The direct-or-abstain ledger in Table 9 reports the candidate and cash outcomes separately. Tables 10, 11, and 12 retain the prototype's concentration and validation evidence.

**Table 9:** Whole-contract direct conversion and cash abstention outcomes for the High Ceiling base and diagnostic books.

| Method            | Candidates | Feasible | Selected | Mean utility |
|-------------------|------------|----------|----------|--------------|
| Cash abstention   | 744        | 744      | 123      | 0.00000      |
| Direct truncation | 744        | 621      | 621      | 0.00376      |

**Table 10:** Prototype contract breadth, concentration, deployed gross, and capacity budget at the frozen account size.

| Config     | Candidates | Active | Top 5 share | Deployed gross | At cap share | Cap budget |
|------------|------------|--------|-------------|----------------|--------------|------------|
| orig       | 49         | 47     | 0.421       | 0.497          | 0.064        | 0.739      |
| orig+VIX   | 54         | 54     | 0.324       | 0.723          | 0.093        | 1.093      |
| larger     | 262        | 243    | 0.233       | 0.594          | 0.078        | 1.160      |
| larger+VIX | 267        | 253    | 0.296       | 0.836          | 0.083        | 1.515      |

**Table 11:** Pass, mixed, fail, and diagnostic outcomes across the locked prototype validation suite.

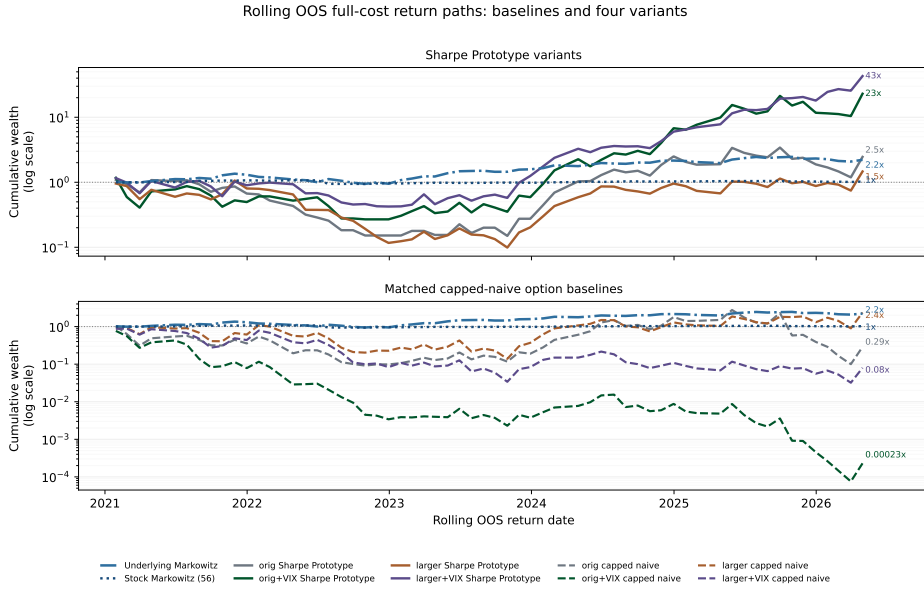
| Config     | Baselines  | Capacity | CPCV net (2018+) | CPCV net claim | MC resampled | MC refit | Rolling OOS | PBO rank | Repriced overlay |
|------------|------------|----------|------------------|----------------|--------------|----------|-------------|----------|------------------|
| orig       | diagnostic | fail     | pass             | pass           | pass         | pass     | pass        | pass     | mixed            |
| orig+VIX   | pass       | pass     | pass             | pass           | pass         | pass     | pass        | pass     | mixed            |
| larger     | mixed      | pass     | pass             | pass           | mixed        | pass     | pass        | mixed    | fail             |
| larger+VIX | pass       | pass     | pass             | pass           | pass         | pass     | pass        | mixed    | fail             |

**Table 12:** Prototype CPCV wealth quantiles, absorption shares, and relative-path evidence for liquid and claim windows.

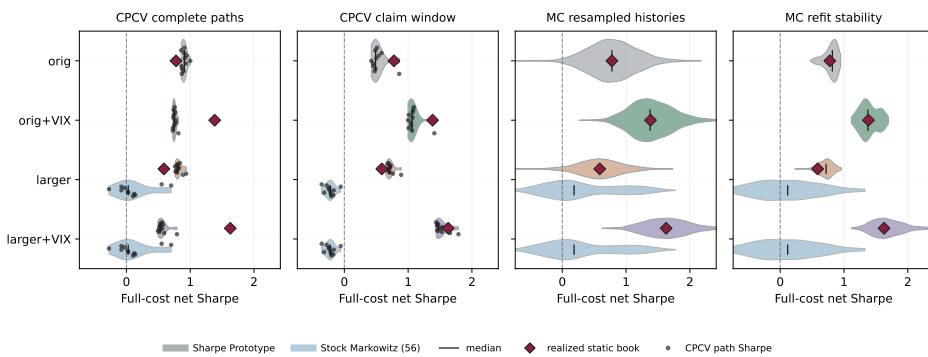
| Config     | Liquid net p05 | Liquid net p50 | Default share | Claim net p05 | Claim net p50 | Rel liquid p05 | Rel claim p05 |
|------------|----------------|----------------|---------------|---------------|---------------|----------------|---------------|
| orig       | 0.860          | 0.908          | 0.000         | 0.437         | 0.490         | -0.561         | -0.125        |
| orig+VIX   | 0.727          | 0.749          | 1.000         | 1.003         | 1.062         | 0.214          | 2.257         |
| larger     | 0.766          | 0.802          | 0.000         | 0.660         | 0.706         | -0.265         | -0.152        |
| larger+VIX | 0.500          | 0.532          | 1.000         | 1.445         | 1.474         | 0.103          | 1.501         |

### A.6. Relocated path figures

The following figures retain the detailed path and distribution views used by the verification suite. Figures 11 and 12 supplement the compact robustness summary in the main text.



**Figure 11:** Rolling out-of-sample full-cost return paths for stock baselines, the Sharpe Prototype universes, and capped naive option books.



**Figure 12:** Distributional validation for the four prototype universes. The shapes summarize locked CPCV, resampling, and refit evidence rather than future returns.

## Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. The repository does not redistribute raw licensed OPRA/Databento data. It instead ships aggregate artifacts in the execution-audit directory and an acquisition summary with request hashes. The audit checks entry-window costs and decision marks for held positions against licensed consolidated quotes. The baseline eight equity-option spread rows use local historical panel CBBO. Broad-name and VIX spread rows without matched historical CBBO use the documented inferred CBBO proxy. The paper labels them as execution-sensitivity evidence. The pipeline versions exact VRO/SOQ settlement files by source hash before VIX option rows enter the reported tables.

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